



SpinFlow

A Physics-Informed Spin Field Framework for
Traffic Phase Inference and Transition Detection

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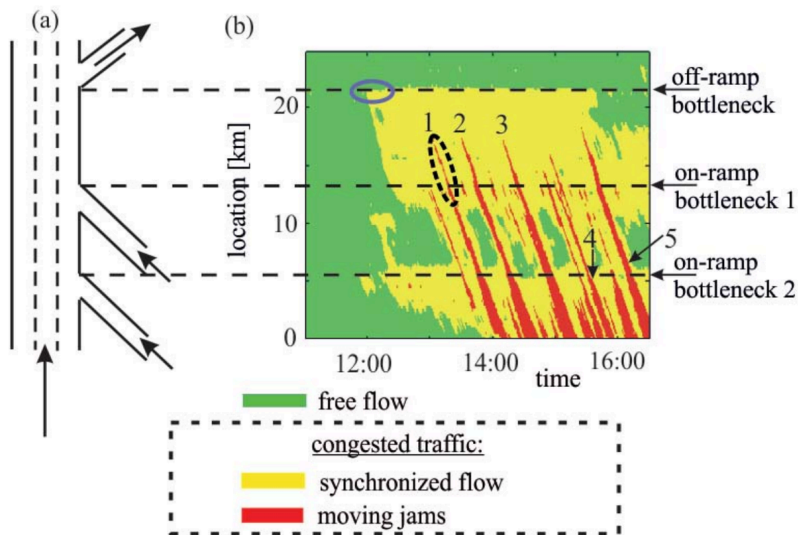
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Presenter: Haopeng Deng • September 17, 2026

arXiv:2605.23306 • github.com/QianyHP/spinflow-traffic

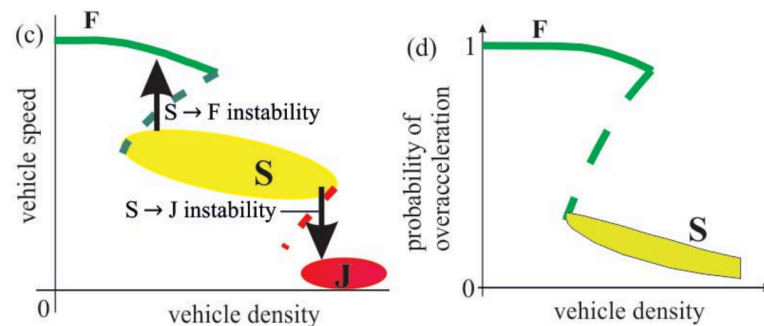
Motivation I: active traffic management reacts too late

02



 **Traffic breakdown is $F \rightarrow S$ transition in metastable free flow** governed by $S \rightarrow F$ instability introduced in three-phase theory.

 **Emergence of wide moving jam is $S \rightarrow J$ transition** governed by traffic flow instability introduced in 1958-1961 by Herman, Gazis et al.



Kerner's three-phase theory reveals a metastable transition regime hard to represent with a single FD [4].

Jams nucleate inside S — creating a hidden transition window before congestion is visible.

Classical models use a single FD and miss S [1-3];

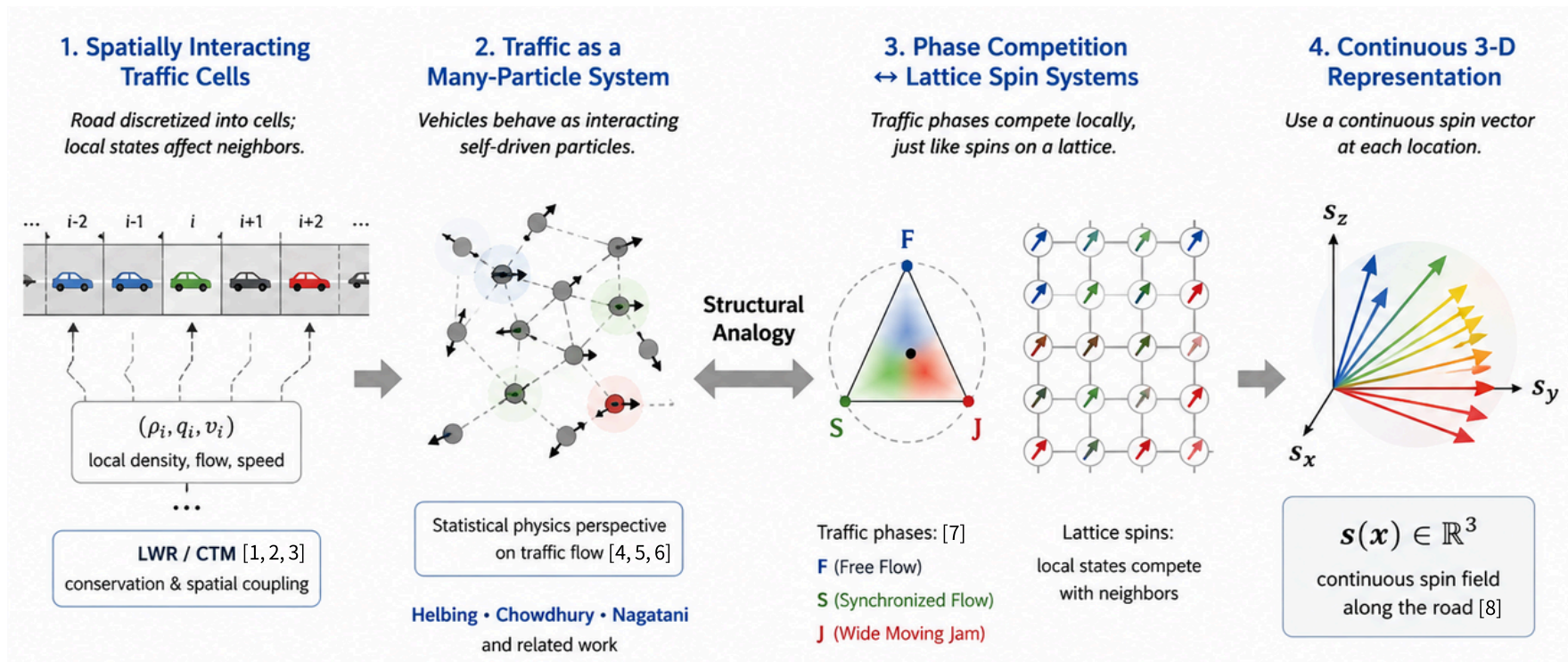
Kerner captures F-S-J transitions but lacks an inferable latent state [4];

Physics-informed ML improves flexibility but may sacrifice interpretability [5].

We need a continuous, interpretable phase state inferred directly from vehicle trajectories data.

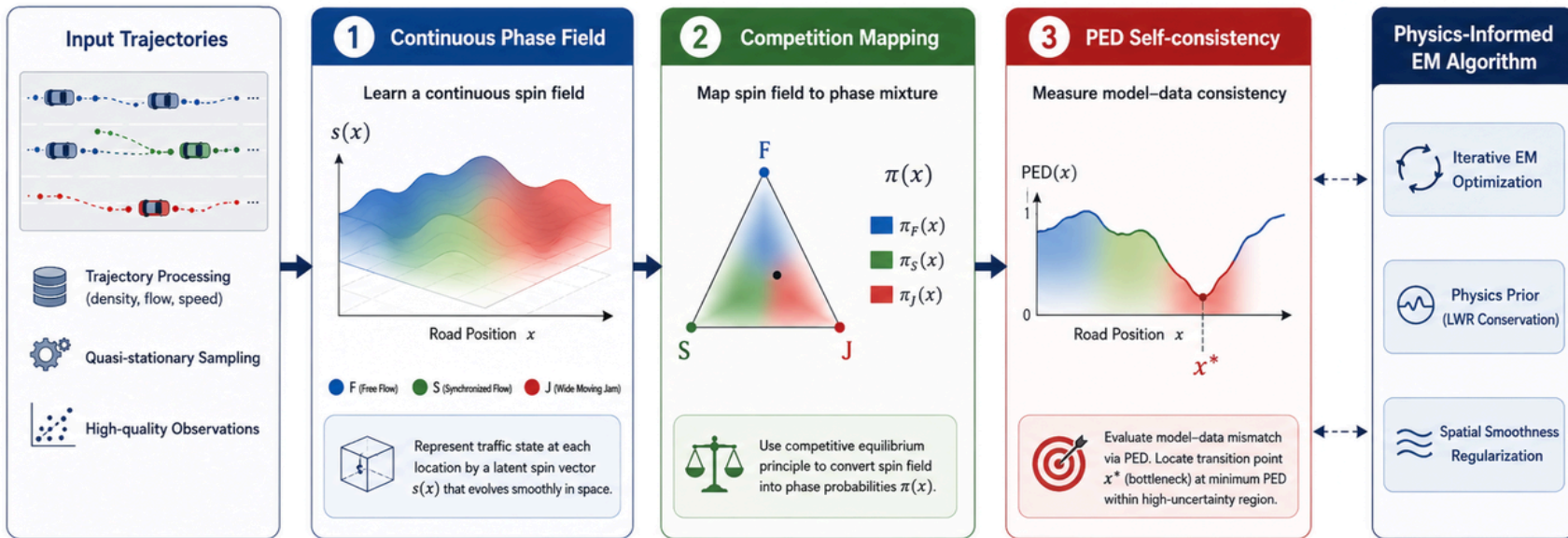
Motivation II: from traffic cells to continuous spin fields

03



[1] Lighthill & Whitham (1955) • [2] Richards (1956) • [3] Daganzo (1994) • [4] Chowdhury et al. (2000)

[5] Helbing (2001) • [6] Nagatani (2002) • [7] Kerner (2019) • [8] Heisenberg (1932)

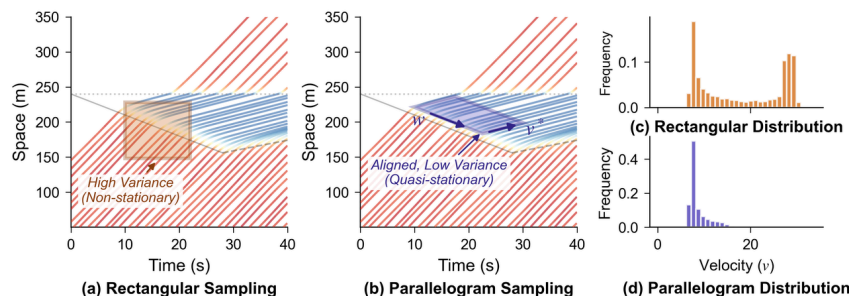


Instead of classifying traffic with hard thresholds, SpinFlow learns a continuous phase field from vehicle trajectories. We then track where the competition between phases becomes inconsistent with the data, allowing us to localize the onset of congestion.

From trajectories to macroscopic states [1]

$$\rho = \frac{\tau}{|\mathcal{A}|}, \quad q = \frac{d}{|\mathcal{A}|}, \quad v = \frac{q}{\rho}$$

Each sampled region yields a spatially anchored state (ρ_p, q_p, v_p, x_p) .



Rectangular vs. parallelogram sampling [2]. Parallelogram keeps trajectories nearly parallel inside.

Quasi-stationary sampling

Parallelograms align with wave propagation: long edge // wave speed w , short edge // target speed v^* . Trajectories inside are nearly parallel $\rightarrow$ the sampled point is quasi-stationary.

Quality-weighted observations

Each sample receives a continuous stability weight w_p from local speed stability scored by

$$CV + NAE \rightarrow \text{adaptive Otsu threshold} \rightarrow w_p.$$

Transition-zone observations remain available for inference.

Output: spatially anchored FD observations $\{(\rho_p, q_p, x_p), w_p\}$, the computational input for prototype EM inversion later on.

Three-phase prototype FD & spatial mixture

06

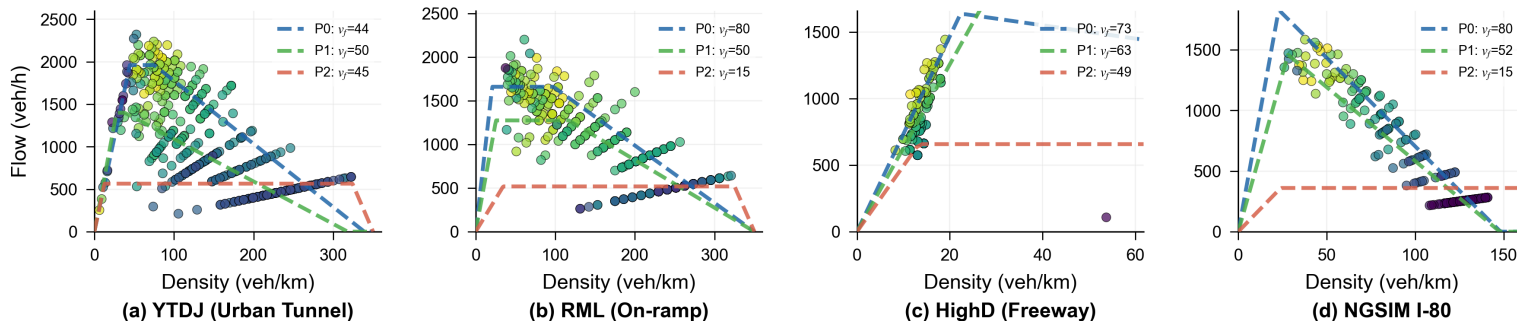
One triangular prototype for each phase $g \in \{F, S, J\}$

$$q_g(\rho) = \min \left\{ v_f^{(g)} \rho, Q^{(g)}, |w^{(g)}| (\rho_{\text{jam}}^{(g)} - \rho) \right\}$$

Parameters $\theta_g = (v_f, w, \rho_{\text{jam}}, Q)$ fitted by weighted least squares from the quality-weighted observations.

Spatial mixture representation: $q(\rho; x) = \sum_g \pi_g(x) q_g(\rho)$

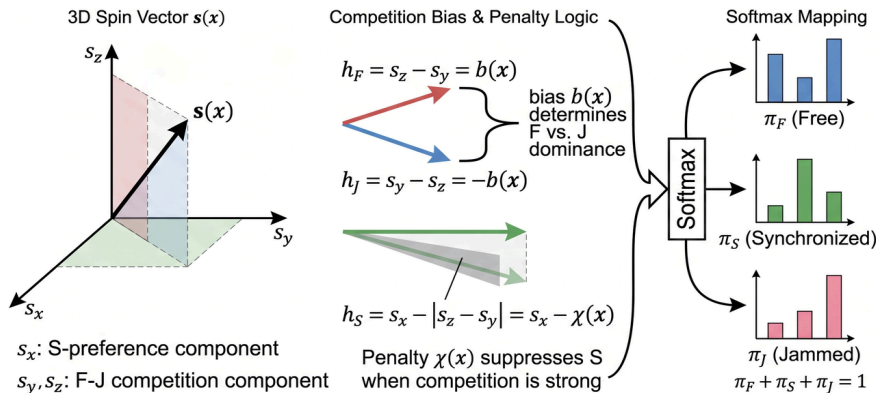
The local proportions $\pi(x)$ are what the latent model must determine.



Fitted prototypes across four scenarios. P0/P1/P2 = F/S/J; scatter: parallelogram-sampled observations; dashed: triangular prototypes.

Spin field & competitive equilibrium mapping

07



Latent spin field at every location

$$\mathbf{s}(x) = (s_x, s_y, s_z) \in \mathbb{R}^3$$

Competition mapping to $\pi(x)$

$$h_F = b, \quad h_J = -b, \quad h_S = s_x - \chi$$

$$b = s_z - s_y, \quad \chi = |b|, \quad \pi_g = \frac{e^{h_g}}{\sum_{g'} e^{h_{g'}}}$$

$\chi = |b|$ is a designed prior: S is penalized unless free and jam are balanced.

Phase Unfolding: magnitude encodes certainty

$\|\mathbf{s}\| \rightarrow 0 \rightarrow$ high-entropy mixture; data is ambiguous or transitional.
 $\|\mathbf{s}\| \gg 1 \rightarrow$ deterministic phase; one phase clearly dominates.

Magnitude is learned, not fixed to unit norm like original Heisenberg model — Spinflow admits ambiguity where data is transitional.

How F-J competition gates S

F and J compete through $b = s_z - s_y$: $b > 0$ favors F, $b < 0$ favors J.

S activates only when $b \approx 0$ — the transition zone where Kerner observed jam nucleation.

E-step: responsibility & kernel aggregation

$$\gamma_{p,g} = \frac{\pi_g \mathcal{N}(q_p; q_g)}{\sum_{g'} \pi_{g'} \mathcal{N}(q_p; q_{g'})}, \quad \pi_{\text{tar},g}(x) = \frac{\sum_p w_p K(x, x_p) \gamma_{p,g}}{\sum_p w_p K(x, x_p)}$$

Soft assignment $\gamma_{p,g}$ aggregated by Gaussian kernel into a continuous field.

Kernel width $\sigma \approx 1.5\Delta x$.

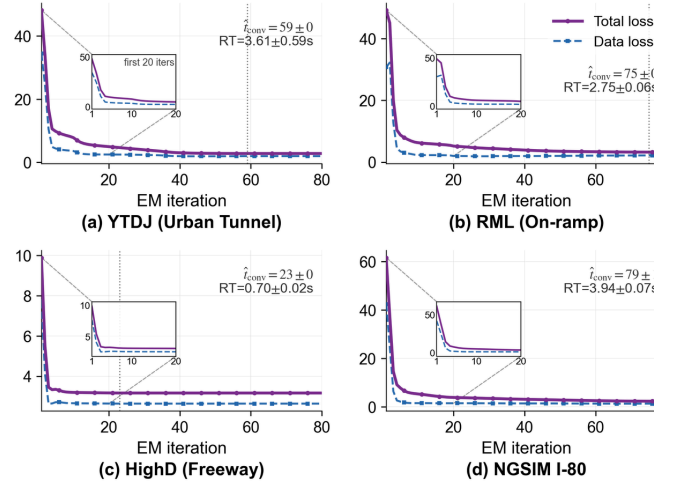
M-step: prototype FD & spin field

$$\theta_g^* = \arg \min_{\theta_g} \sum_p \gamma_{p,g} w_p (q_g(\rho_p) - q_p)^2$$

$$\mathbf{s}^* = \arg \min_{\mathbf{s}} \int \left[- \sum_g \pi_{\text{tar},g} \log \pi_g + \lambda_{\text{phys}} \Delta \mathcal{R}^2 + \lambda_{\text{smooth}} \|\partial_x \mathbf{s}\|^2 \right] dx$$

Prototype FD by weighted least squares; spin field by gradient descent on cross-entropy plus physics and smoothness terms.

Convergence & efficiency



Total loss (solid) and data loss (dashed) vs. EM iteration. t_{conv} and runtime annotated. 23–79 iterations to convergence, 0.7–3.9 s CPU per scenario.

Phase Equilibrium Degree (PED) & transition detection

09

① Measure self-consistency

$$\text{PED}(x) = e^{-D_{\text{KL}}(\pi_{\text{tar}} \parallel \pi)}$$

PED → 1: the inferred mixture explains the local evidence.

PED ↓: stronger model–data inconsistency.



② Gate on phase coexistence

$$\Omega_Q = \{x \mid H(x) \geq \ln 2\}$$

Search only where at least two phases coexist with non-trivial weight.
Pure free flow or pure jam is already deterministic.



③ Locate the transition

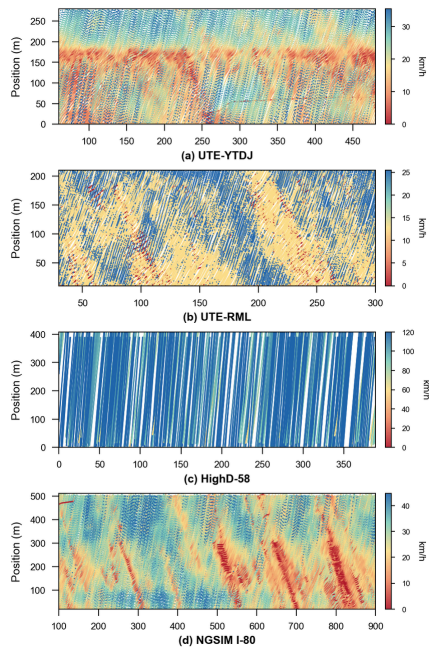
$$x^* = \arg \min_{x \in \Omega_Q} \text{PED}(x)$$

Within the coexistence region, select the location with minimum PED.

The transition site is **not necessarily the densest point** — it is where model–data equilibrium breaks most severely within the coexistence region.

Evaluation across four diverse real-world road segments

10



Scenario	Road setting & documented feature	Sensing	N_{veh}
UTE-YTDJ [1]	Urban tunnel, 10→6 lane drop	Drone	1,072
UTE-RML [1]	Urban arterial, on-ramp merge	Drone	555
HighD-58 [2]	Freeway, no active bottleneck	Drone	373
NGSIM I-80 [3]	Freeway, downstream on-ramp boundary	Roadside video	1,634

Cross-source acquisition

Drone tracking (UTE, HighD) versus roadside video (NGSIM I-80).

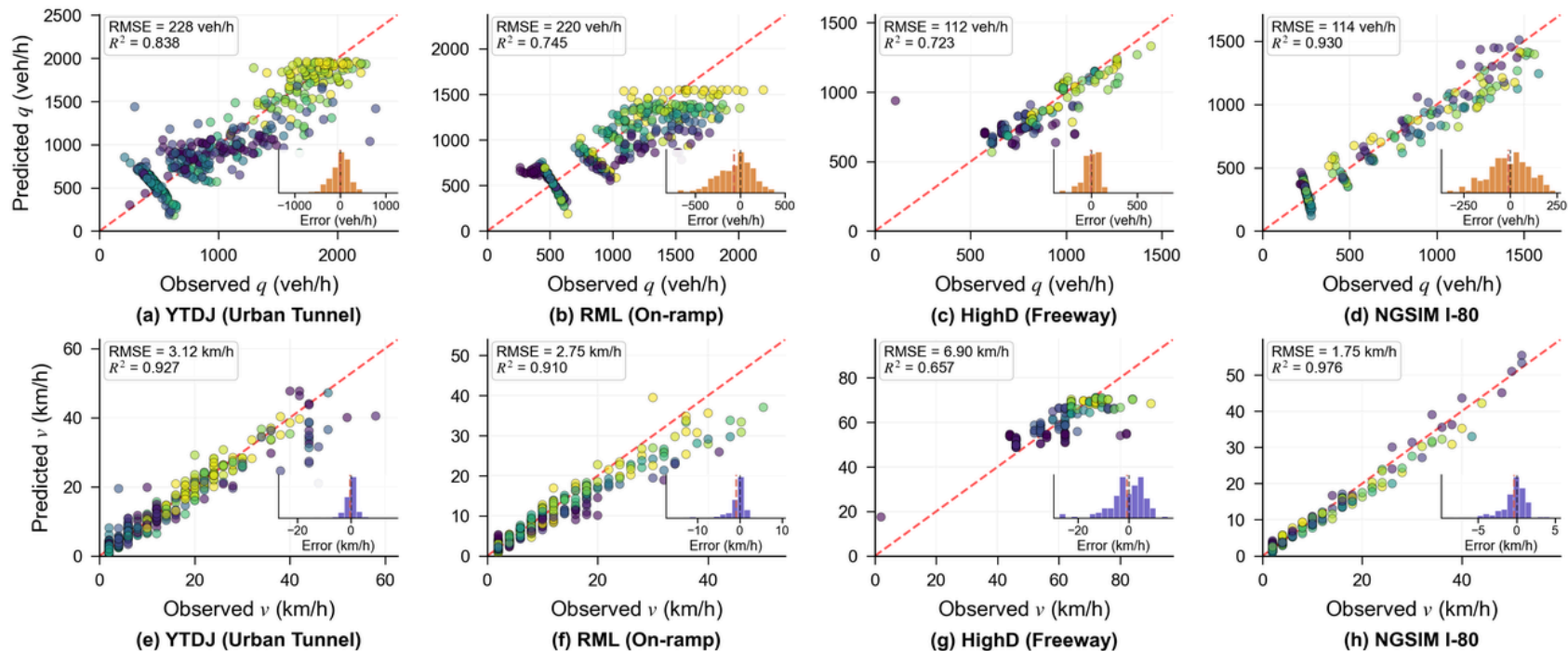
10–30 FPS • 300–800 s per segment • 161–564 FD points

A deliberately diverse testbed spanning road types, bottleneck conditions, and sensing modalities.

Forward consistency: reconstructing observations

11

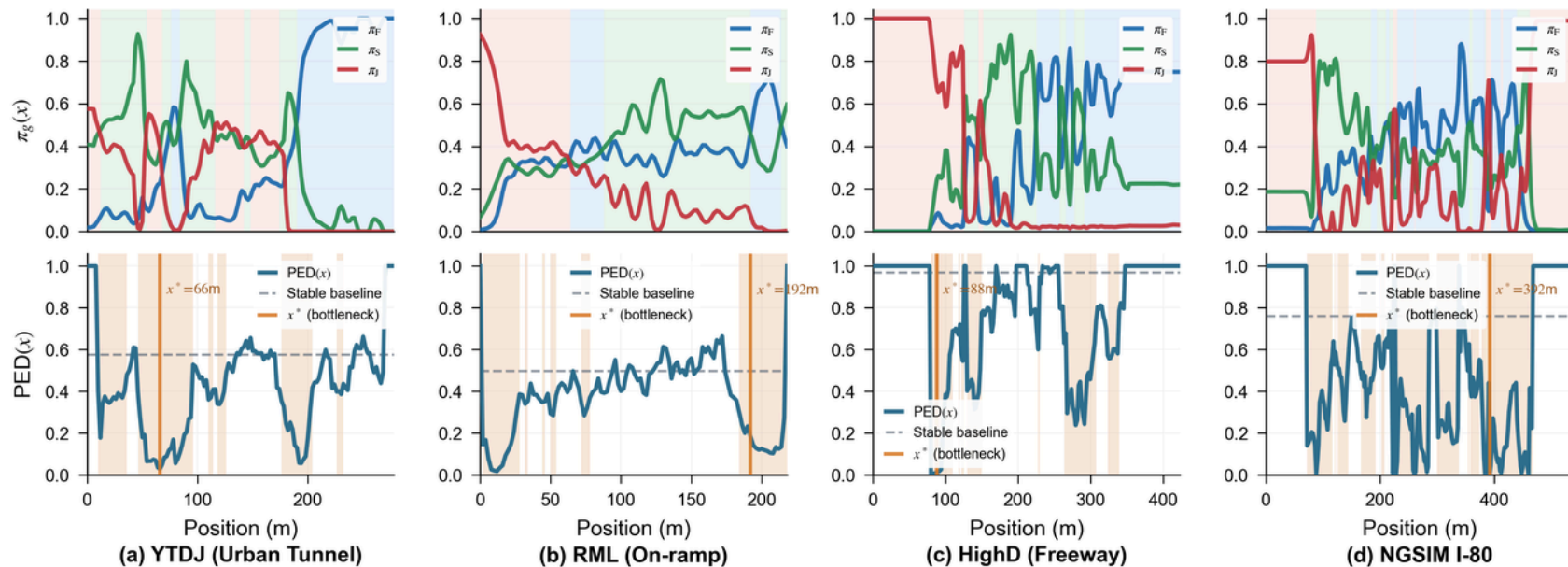
Do the inferred phases reconstruct the observed FD points? Color encodes spatial position x .



Observed vs. reconstructed flow (top) and speed (bottom) for four scenarios. Insets: residual histograms.

Phase inference & bottleneck localization

12



Top: phase weights $\pi_g(x)$ (blue = free, green = synchronized, pink = jam). Bottom: $PED(x)$ with detected transition site x^* .

Detected sites match documented geometry: YTDJ 66 m (upstream lane-drop), RML 192 m (on-ramp mixing edge), NGSIM 392 m (downstream on-ramp boundary), HighD 88 m (mild).

PED drop at x^* : 94.9–100% relative to stable baseline. Entropy at x^* exceeds $\ln 2$, confirming mixed-phase coexistence.

Ablation: removing the competition penalty

$$\text{No-Comp: } h_S = s_x \quad (\text{set } \chi = |b| = 0)$$

Without the F–J balance penalty, synchronized flow is no longer suppressed when free and jam are imbalanced, and gets flattened everywhere.

YTDJ
+5.7%

RML
+6.5%

HighD
+21.5%

NGSIM
+24.3%

RMSE_q increase, 5 seeds, $p < 0.05$ paired t-test. Sparser scenarios suffer more.

Comparison with baselines

Method (paradigm)	$\pi(x)$	PED	R_q^2 (range)	CPU (s)
SpinFlow (ours) • physics-informed	Yes	Yes	0.72 – 0.94	0.7 – 3.9
PWA-CTM [34] • white-box	No	No	0.36–0.57	<1 (est.)
VBGMM + KDE [35] • grey-box	uncalibrated	No	0.69–0.82	~15 (est.)
PI-DeepONet [36] • black-box	No	No	0.82 on RML	6–35 (~9×)

01

Make phase competition computable

Continuous $s(x)$ provides a spatially varying, interpretable representation of F–S–J phase competition.

02

Make equilibrium breaking detectable

PED provides a label-free signal for locating where model–data equilibrium breaks.

03

Make transition localization physically grounded

Detected sites align with documented bottleneck geometry across four diverse real-world scenarios.



Thank you

Questions & discussion welcome.

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I'm actively seeking PhD opportunities for Fall 2027.

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arXiv:2605.23306 • github.com/QianyHP/spinflow-traffic



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